University of Al-Ameed
College of Medicine
Department of Anatomy

Lecture 4

ANATOMY OF THE NECK

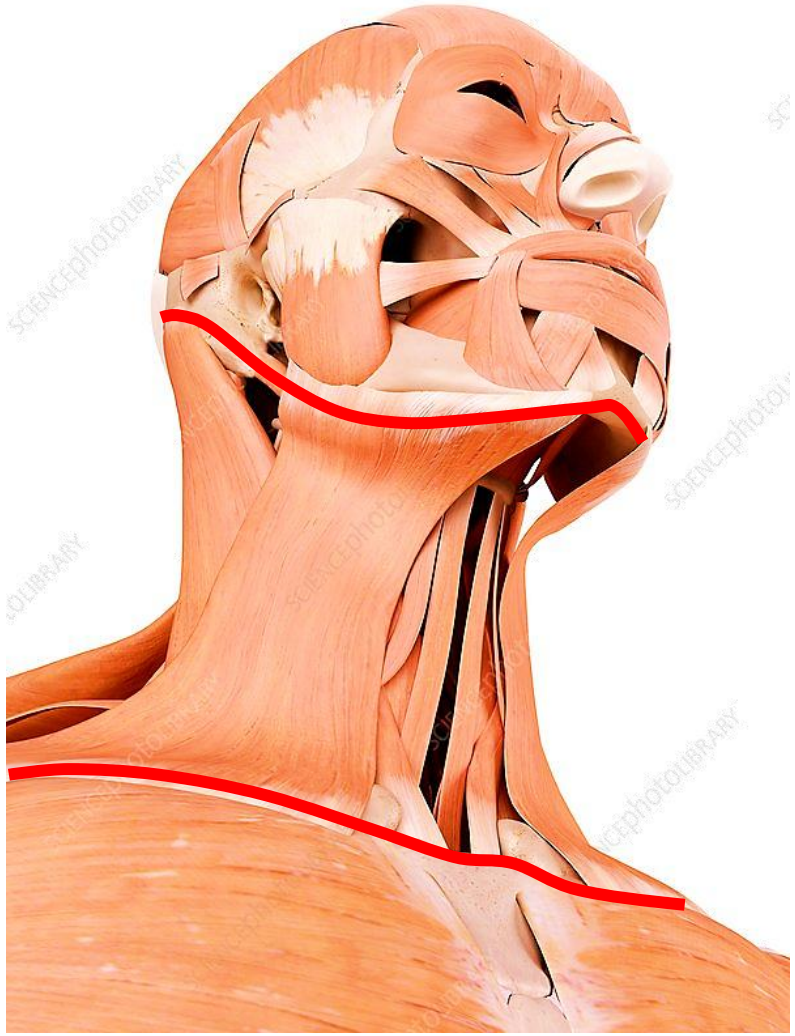
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Learning Objectives

- Define the neck and list its boundaries.
- Classify the fascial compartments of the neck.
- Describe the cervical fascia.
- Demonstrate the superficial cervical veins.
- Illustrate the cervical triangles.
- Describe the muscles of the neck.

THE NECK

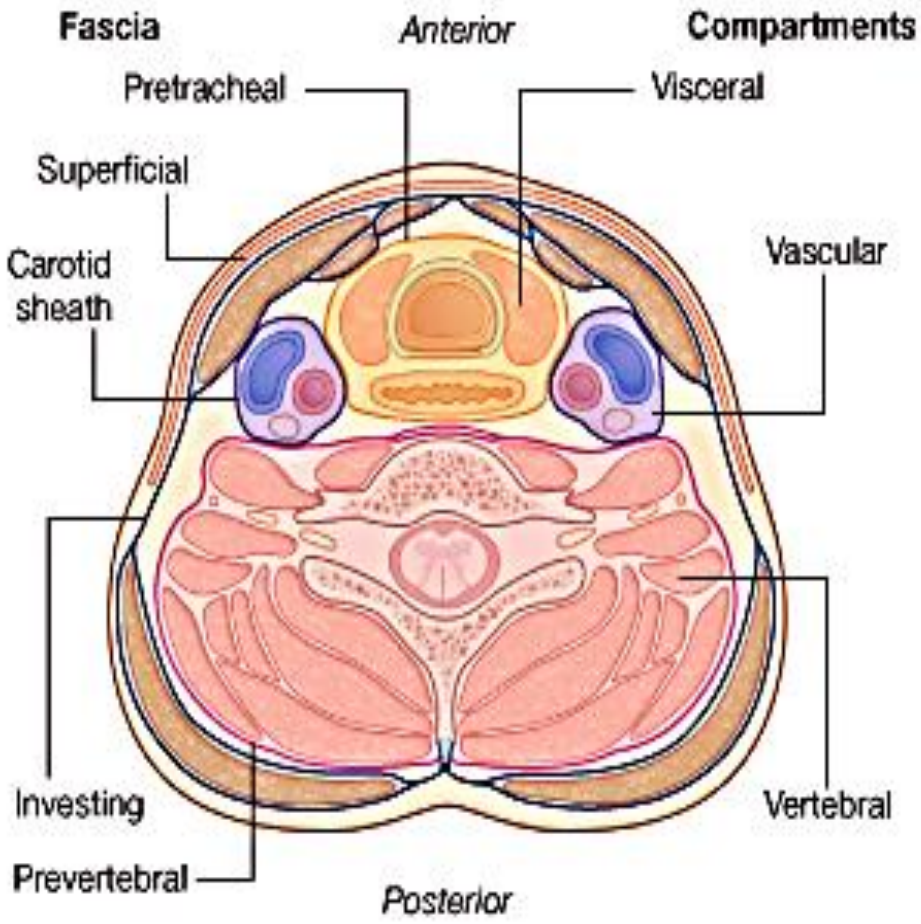
- The neck is a tube providing continuity from the head to the trunk.
- It extends anteriorly from the lower border of the mandible to the upper surface of the manubrium of sternum, and posteriorly from the superior nuchal line on the occipital bone of the skull to the intervertebral disc between the C7 and T1 vertebrae.



Compartments of the Neck

- Within the tube four compartments provide longitudinal organization:

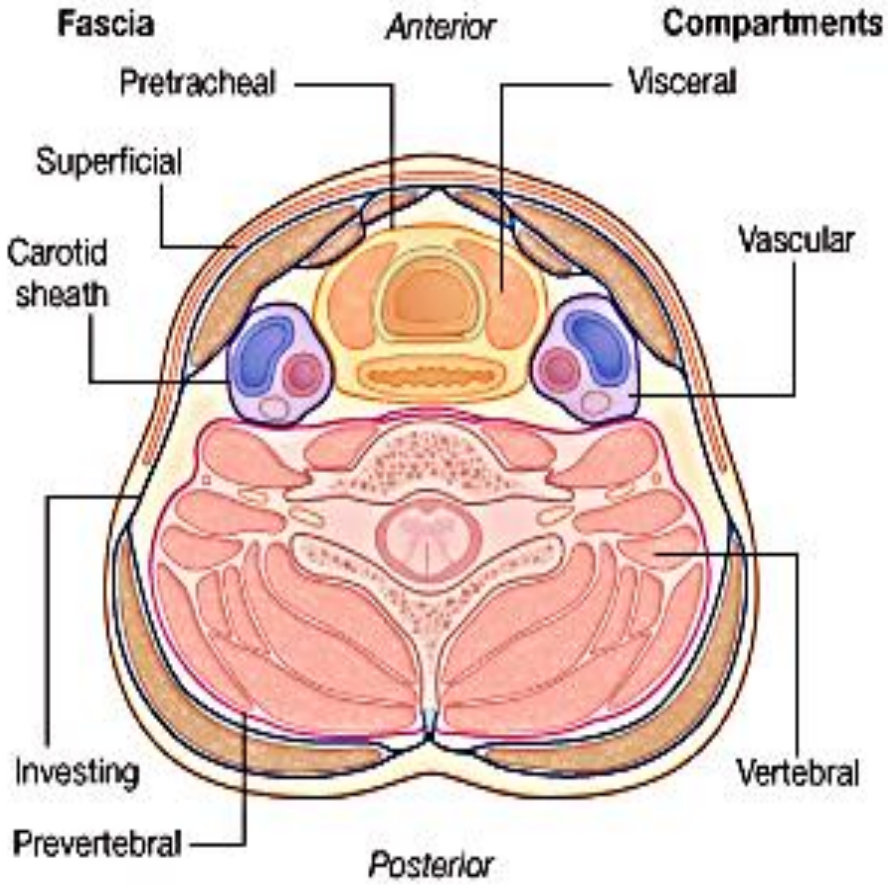
1. **Visceral compartment** is anterior and contains parts of the digestive and respiratory systems, and several endocrine glands
2. **Vertebral compartment** is posterior and contains the cervical vertebrae, spinal cord, cervical nerves, and muscles associated with the vertebral column.
3. Two **vascular compartments** are lateral and contain the major blood vessels and the vagus nerve.



Fascia of the Neck

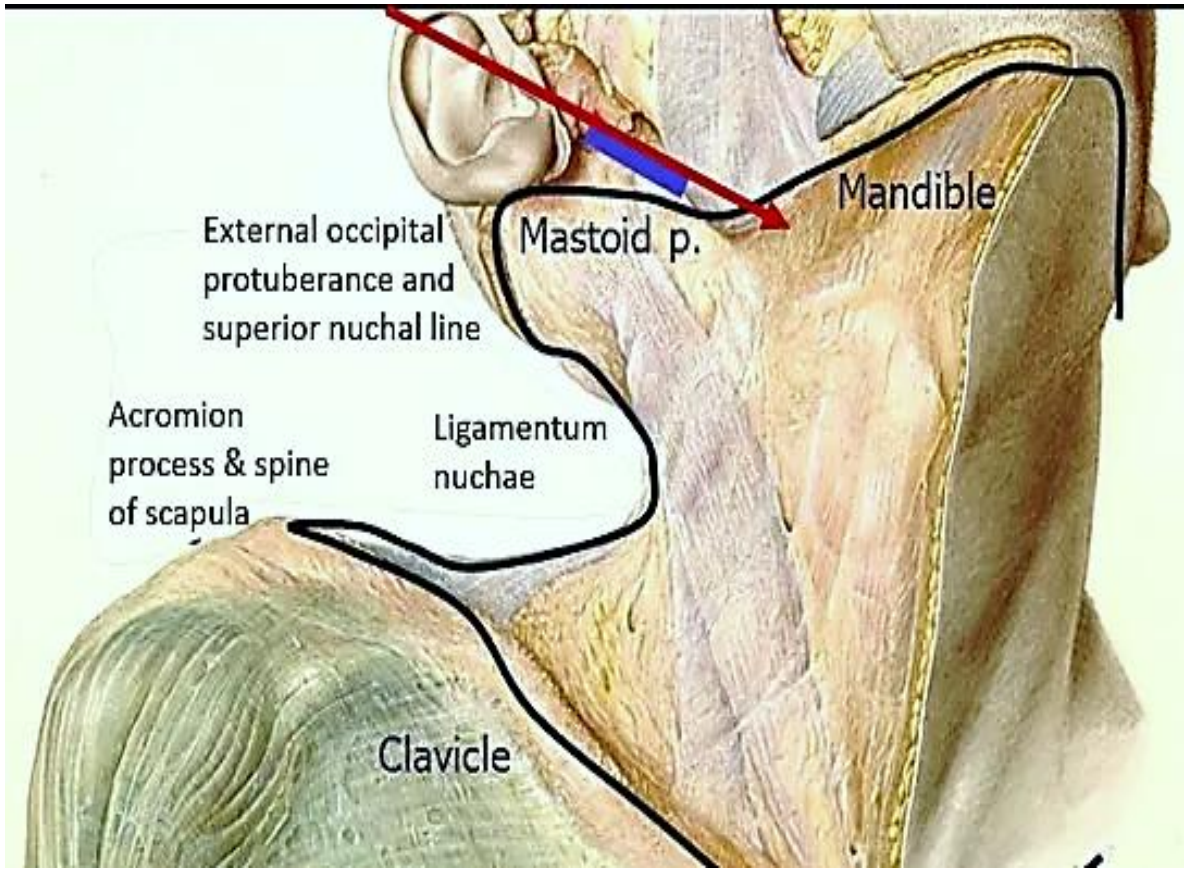
- The superficial fascia in the neck contains a thin sheet of muscle (Platysma).
- The deep cervical fascia is organized into several distinct layers:

A. Investing layer
B. Pre-vertebral layer
C. Pre-tracheal layer
D. Carotid sheaths



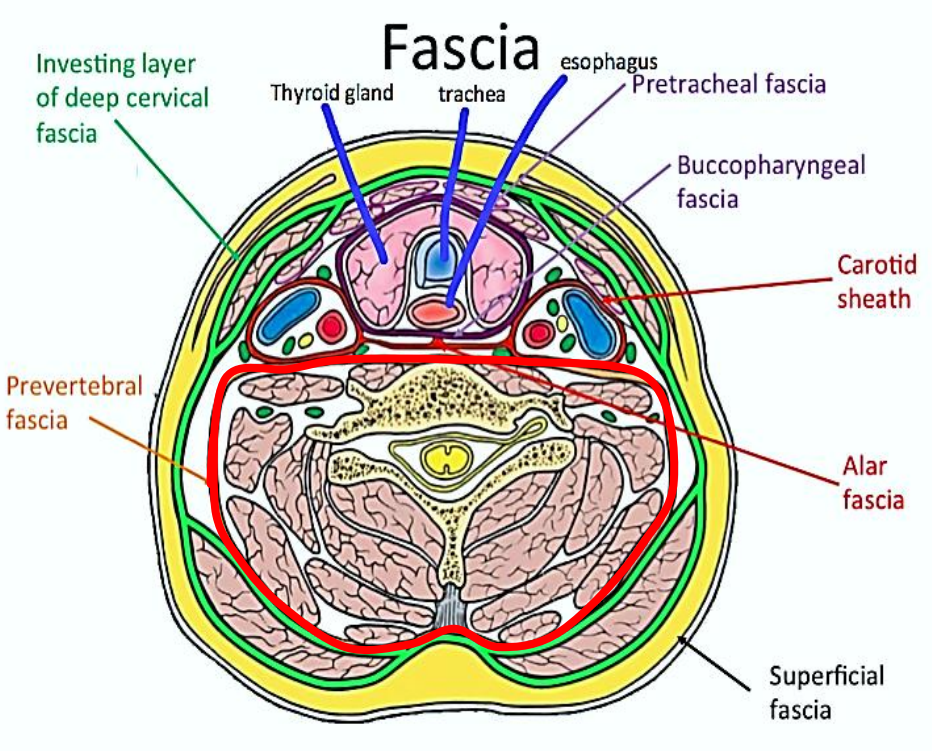
Investing Layer

- It **completely** surrounds the neck.
- Attaching **posteriorly** to the ligamentum nuchae and the spinous process of the C7 vertebra.
- **Anteriorly**, surrounds the infrahyoid muscles.
- It is attached **superiorly** to the external occipital protuberance and the superior nuchal line; **laterally** to the mastoid process and zygomatic arch; and **inferiorly** to the spine of the scapula, the acromion, the clavicle, and the manubrium of sternum.



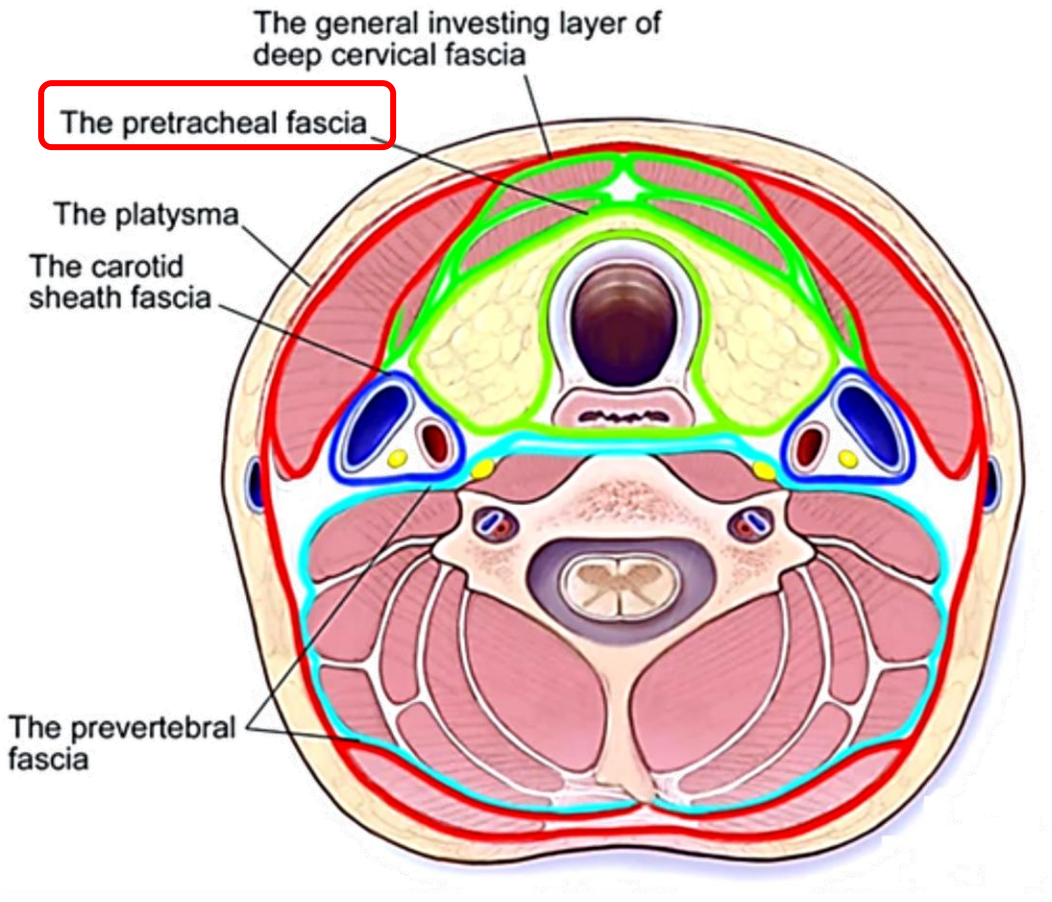
Prevertebral layer

- It is a **cylindrical** layer of fascia that surrounds the vertebral column and the muscles associated with it.
- Muscles in this group include the prevertebral muscles, the anterior, middle, and posterior scalene muscles, and the deep muscles of the back.
- It attaches **superiorly** to the base of skull, and extends **inferolaterally** to surround the brachial plexus and subclavian artery as **Axillary Sheath**.



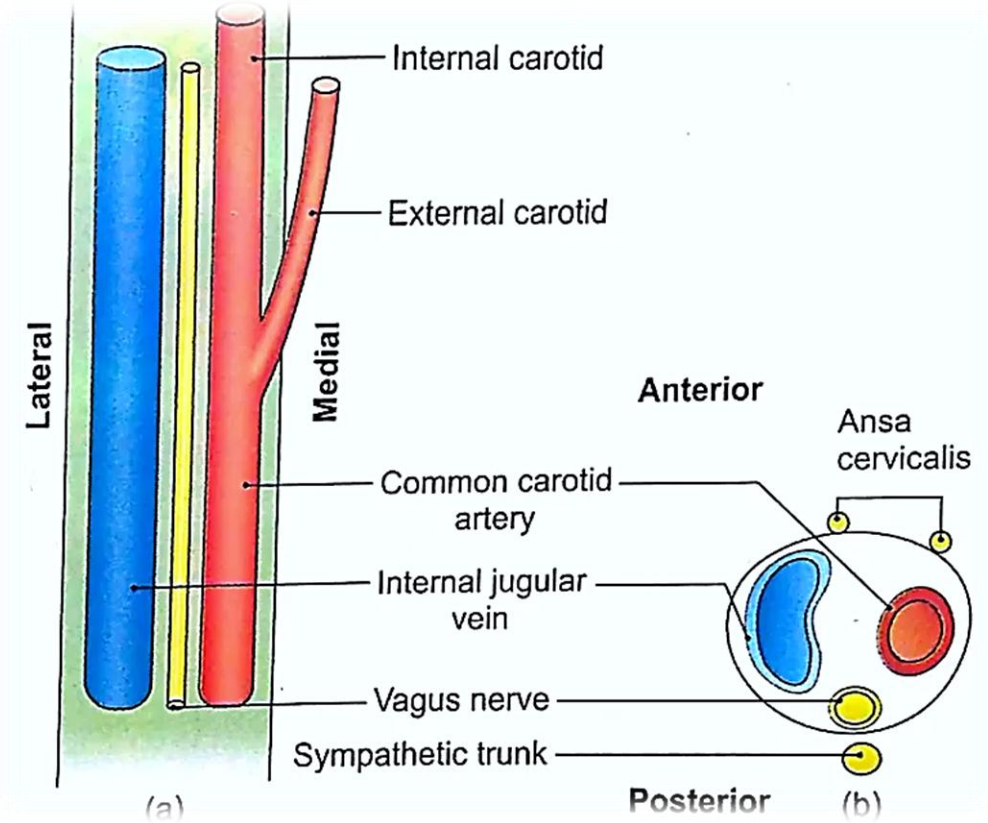
Pretracheal layer

- It consists of a collection of fasciae that surround the trachea, esophagus, and thyroid gland.
- It begins superiorly at the hyoid bone and ends inferiorly in the upper thoracic cavity.



Carotid Sheath

- Each carotid sheath is a column of fascia that surrounds the **common carotid artery**, the **internal carotid artery**, the **internal jugular vein**, and the **vagus nerve** as these structures pass through the neck.



Superficial Venous Drainage

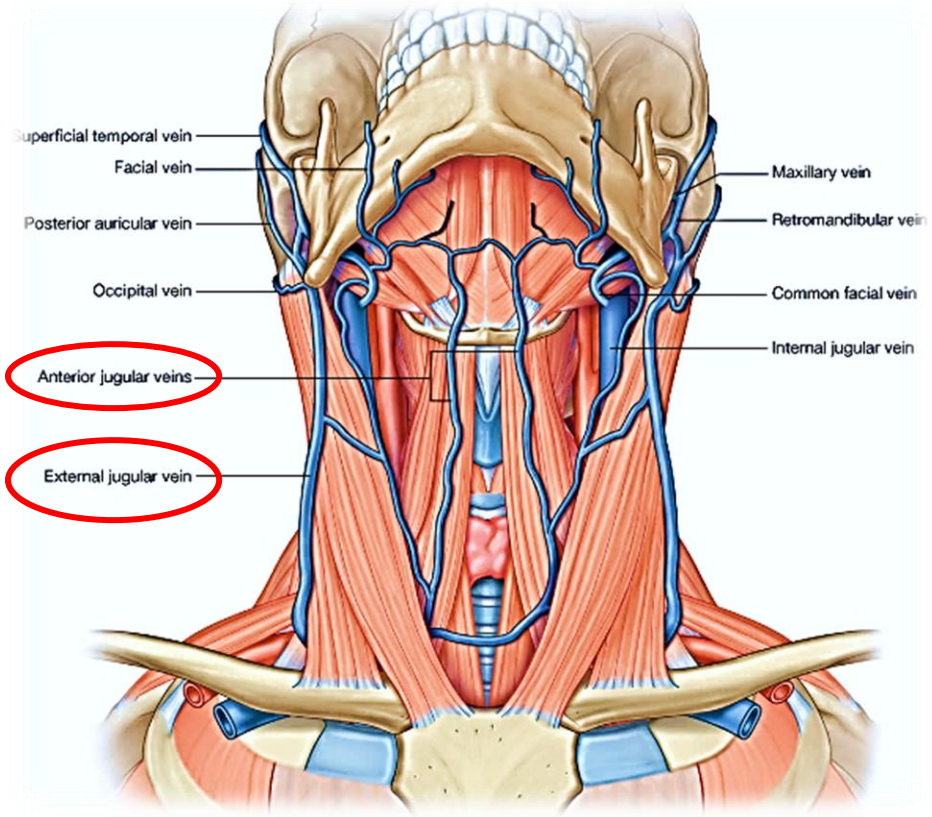
- **External Jugular Veins**

It is formed posterior to the angle of mandible as the **posterior auricular vein** and the **retromandibular vein** join.

- **Anterior Jugular Veins**

Paired venous channels draining the anterior aspect of the neck. They begin above hyoid bone and extends downward on either side of the midline

Both veins pierce the investing layer of cervical fascia to enter the subclavian vein inferiorly.



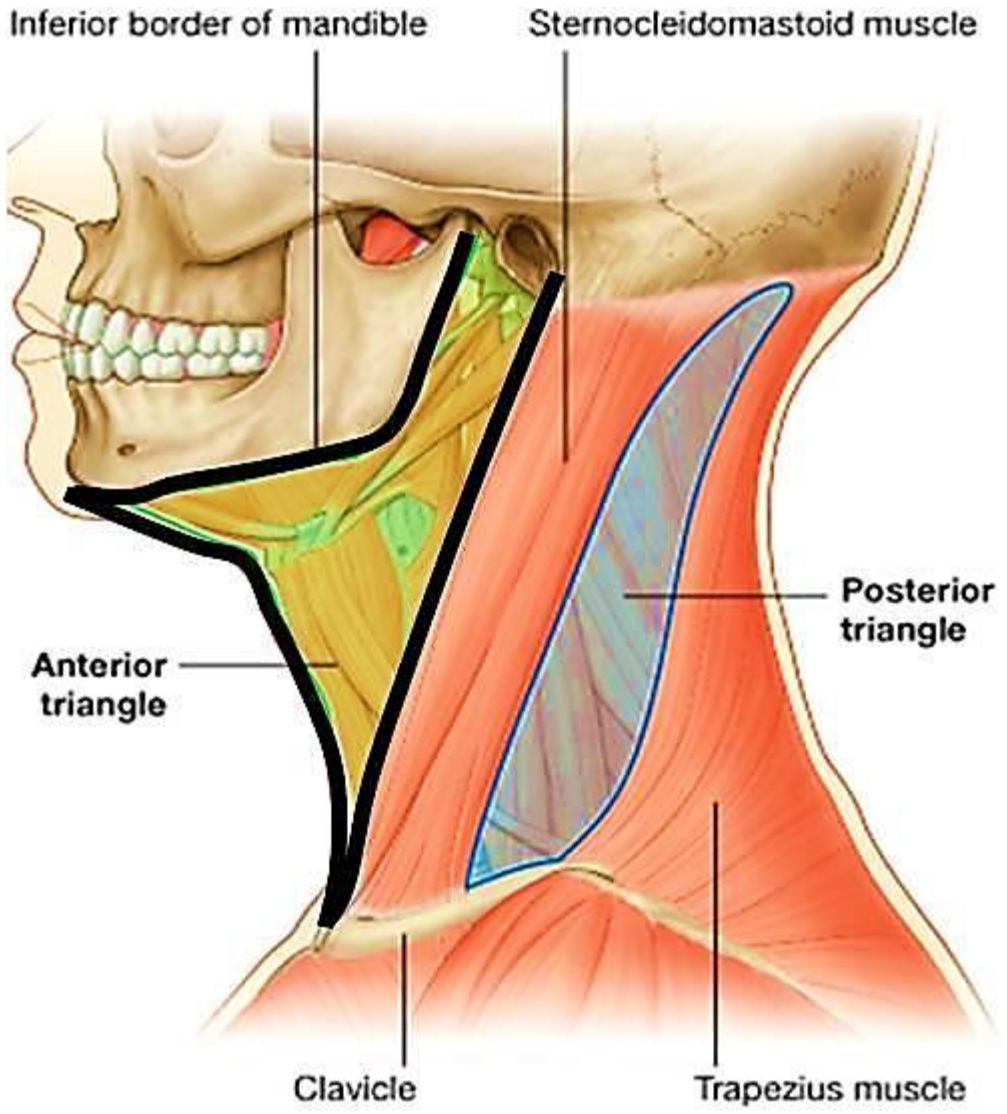
Triangles of the Neck

A. Anterior triangle

Sternocleidomastoid, inferior border of mandible and midline of the neck.

B. Posterior triangle

Sternocleidomastoid, trapezius and middle third of clavicle.



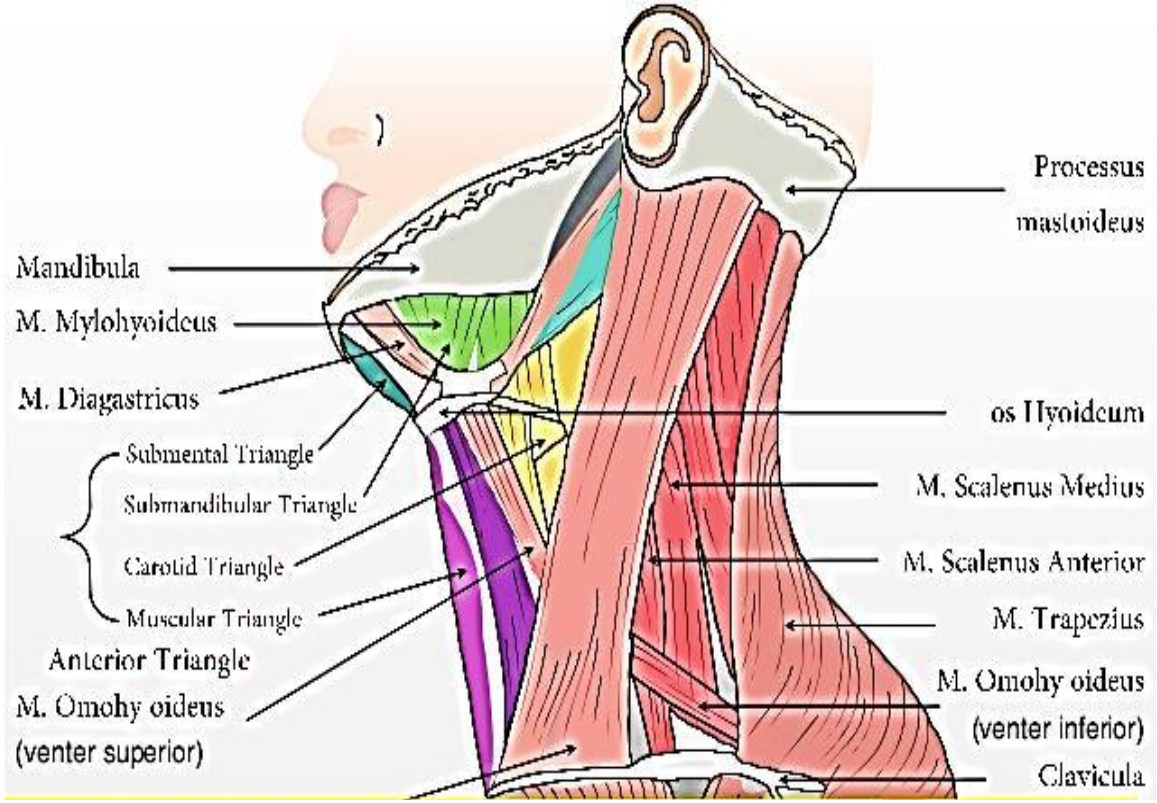
Anterior Triangle of the Neck

A. Submental

B. Submandibular

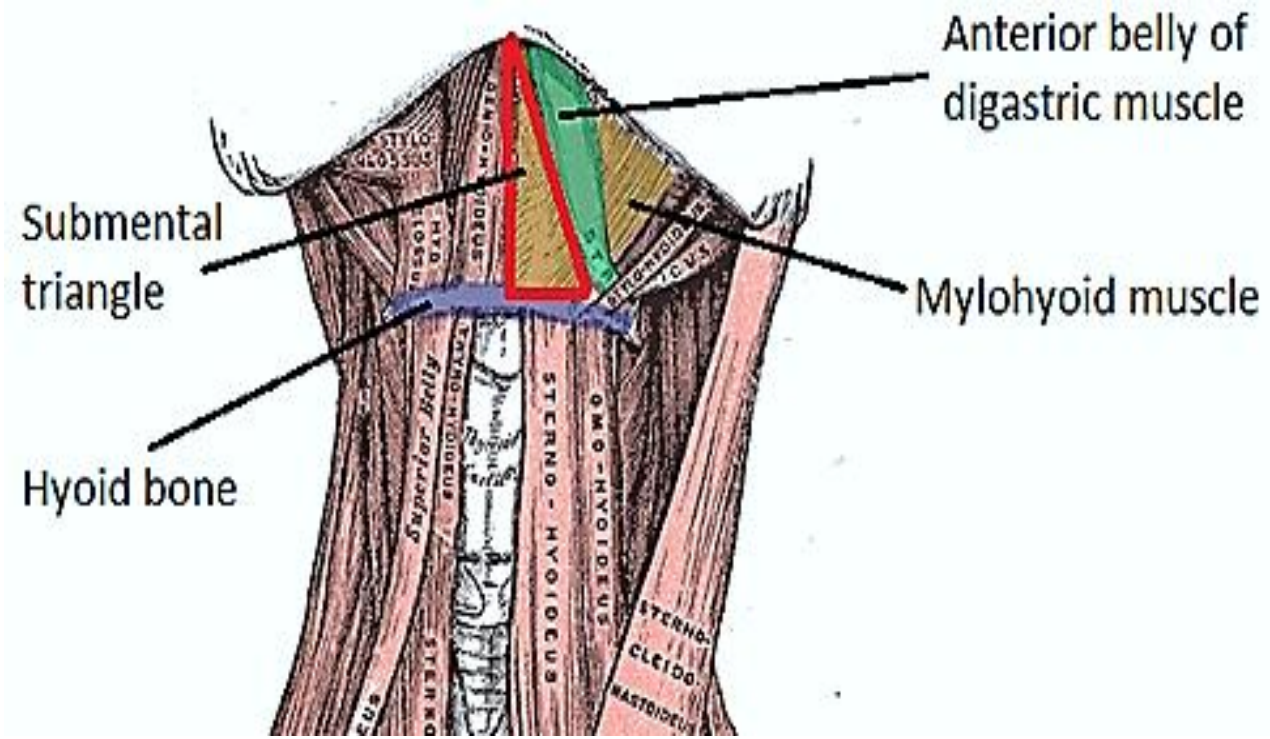
C. Carotid

D. Muscular



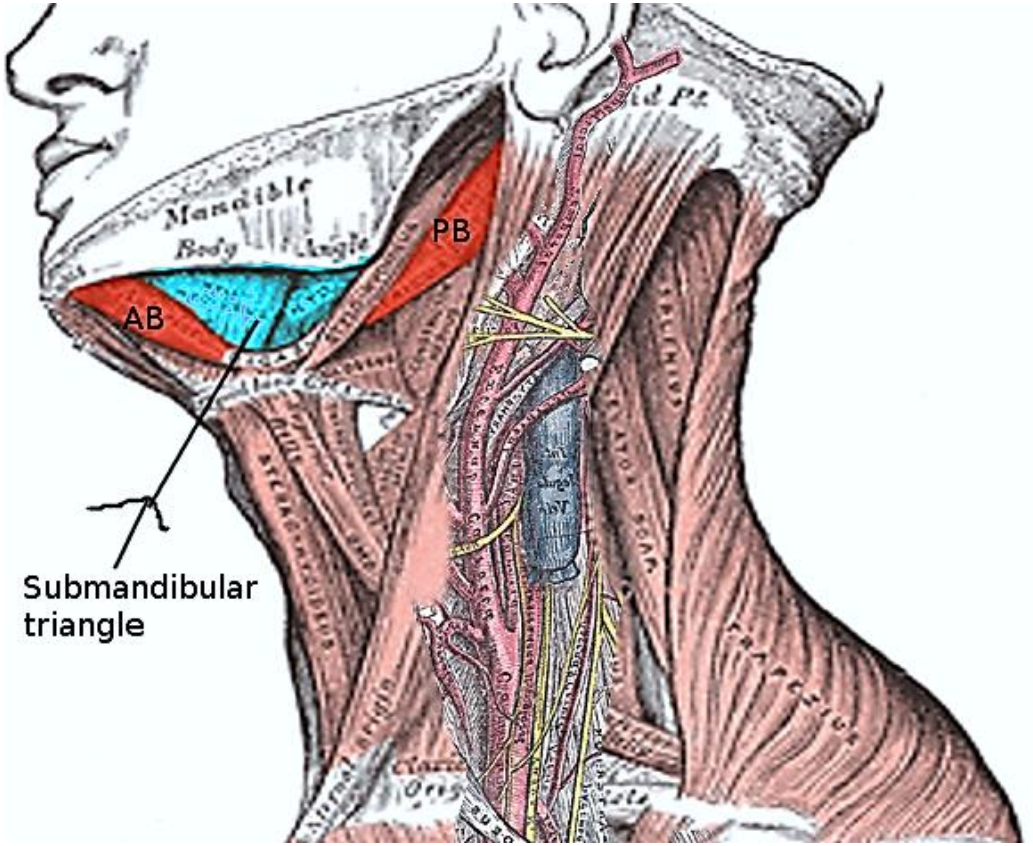
Submental Triangle

- It is outlined by the hyoid bone inferiorly, the anterior belly of the digastric muscle laterally, and the midline.
- It contains the submental lymph nodes and anterior jugular veins.



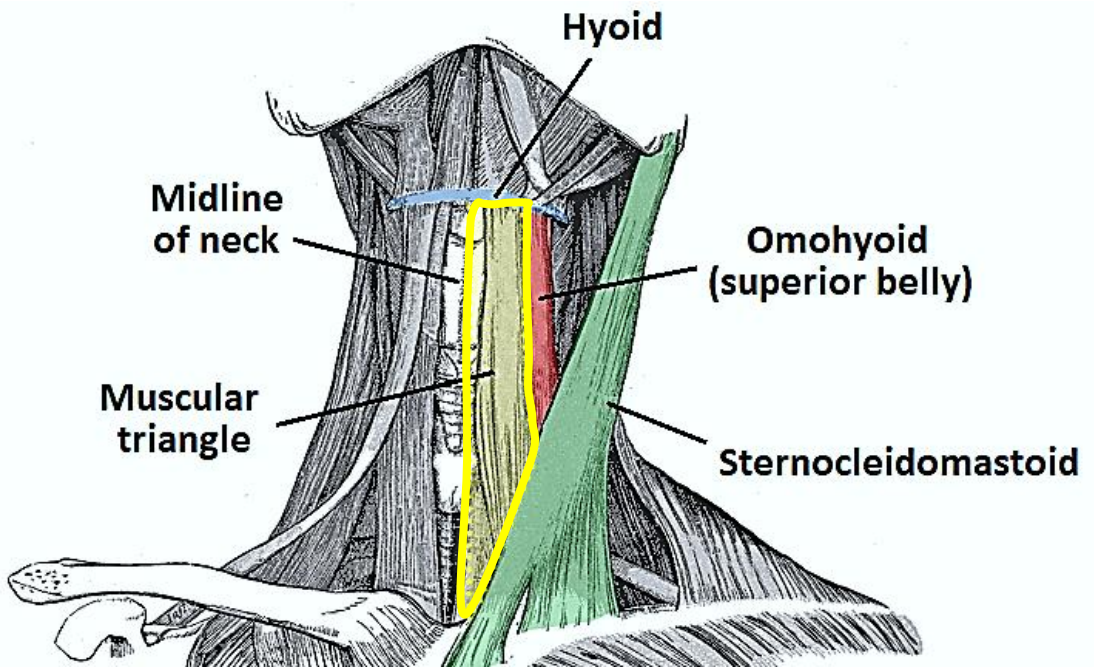
Submandibular Triangle

- It is outlined by the inferior border of the mandible superiorly and the anterior and posterior bellies of the digastric muscle inferiorly.
- It contains the submandibular salivary gland, facial blood vessels, and submandibular lymph nodes.



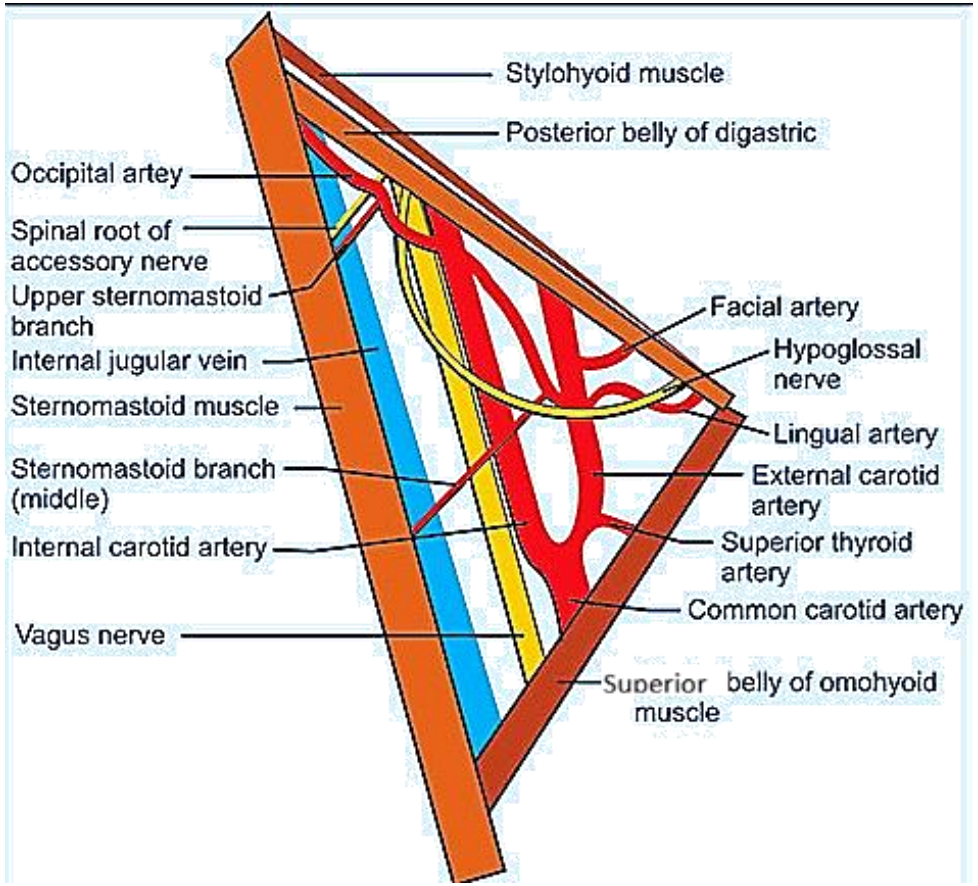
Muscular Triangle

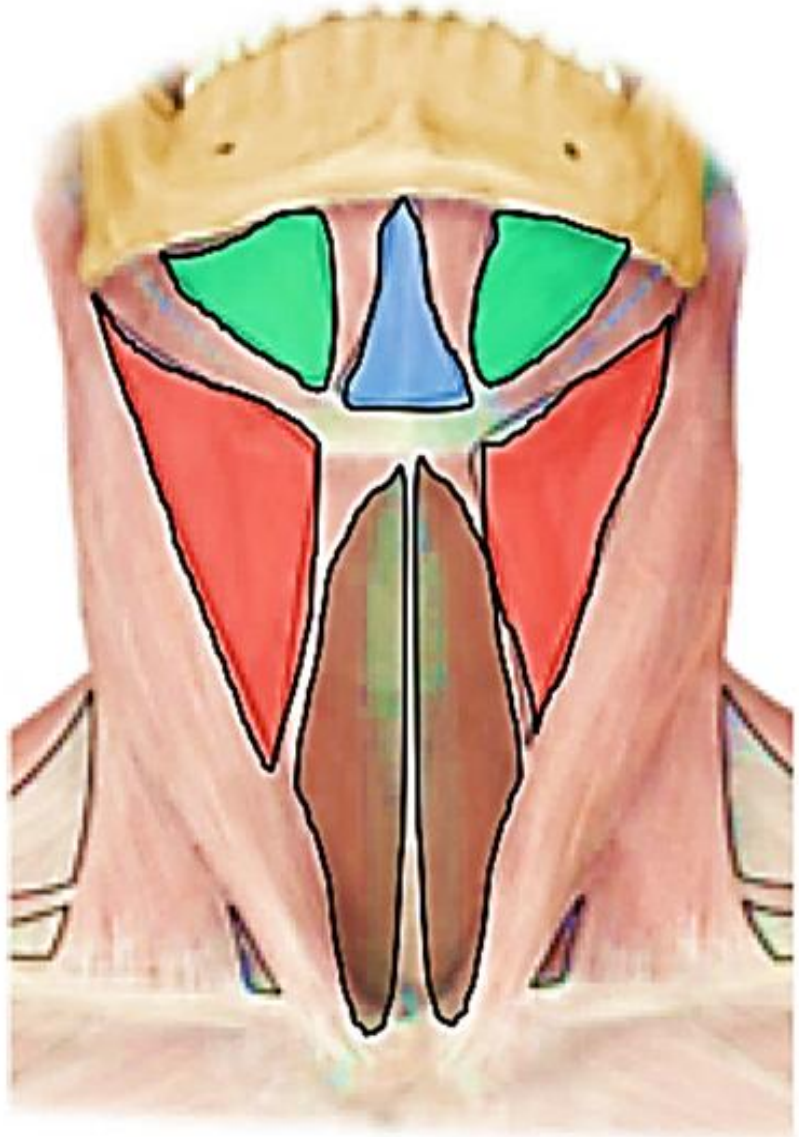
- It is outlined by the hyoid bone superiorly, the superior belly of the omohyoid muscle, and the anterior border of the sternocleidomastoid muscle laterally, and the midline.
- It contains the infrahyoid muscles, pharynx, larynx and thyroid and parathyroid glands.



Carotid Triangle

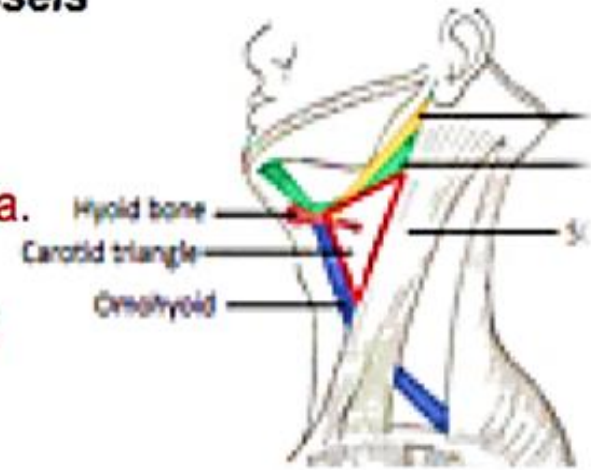
- It is outlined by the superior belly of the omohyoid muscle **anteroinferiorly**, the stylohyoid muscle and posterior belly of the digastric **superiorly**, and the anterior border of the sternocleidomastoid muscle **posteriorly**.
- It contains the carotid sheath that contains the carotid artery, internal jugular vein and vagus nerve, in addition to glossopharyngeal nerve, hypoglossal nerve and deep cervical lymph nodes.





-Muscular triangle

-infrahyoid mm.
-thyroid gland
-contains no vessels



-Carotid triangle

-common carotid a.
-carotid sinus
-internal jugular v.
-CNX and CNXII

-Submandibular (digastric) triangle

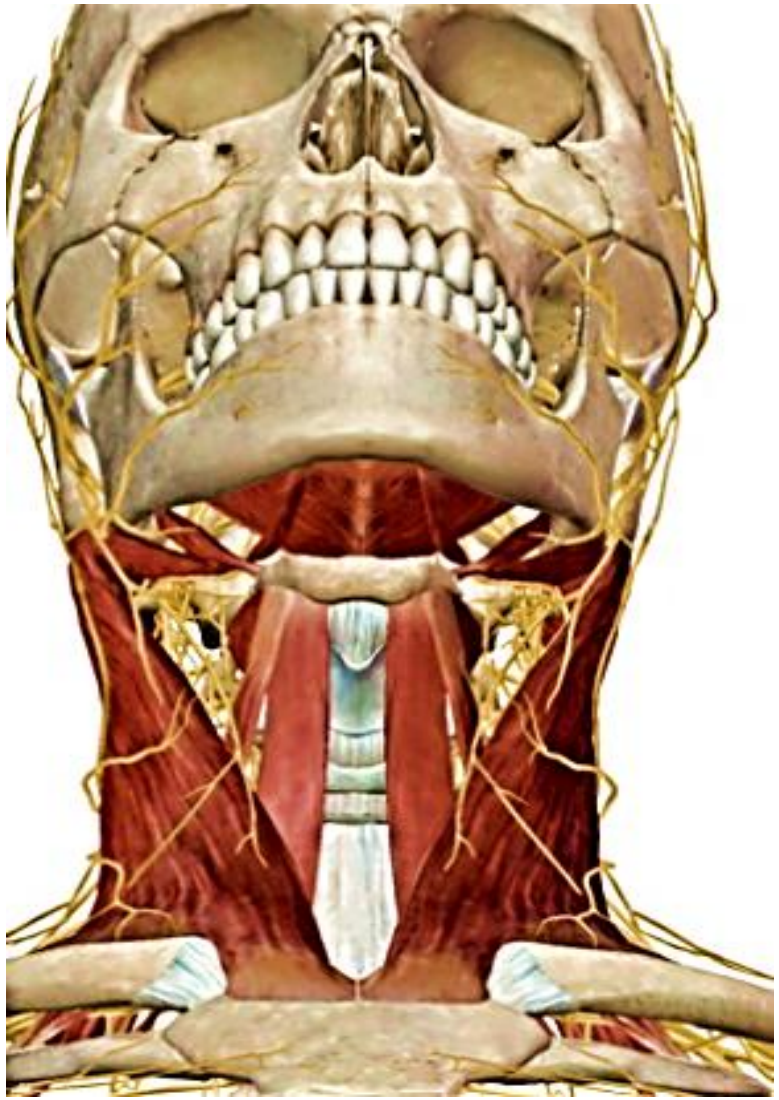
-submandibular gland
-submandibular lymph nodes
-facial a. and v.

-Submental (suprahyoid) triangle

-lymph nodes

Muscles of the neck

- The muscles of the neck are responsible for head movements, stabilizing the upper region of the body, assisting in swallowing, helping to elevate the rib cage during inhalation, and more..



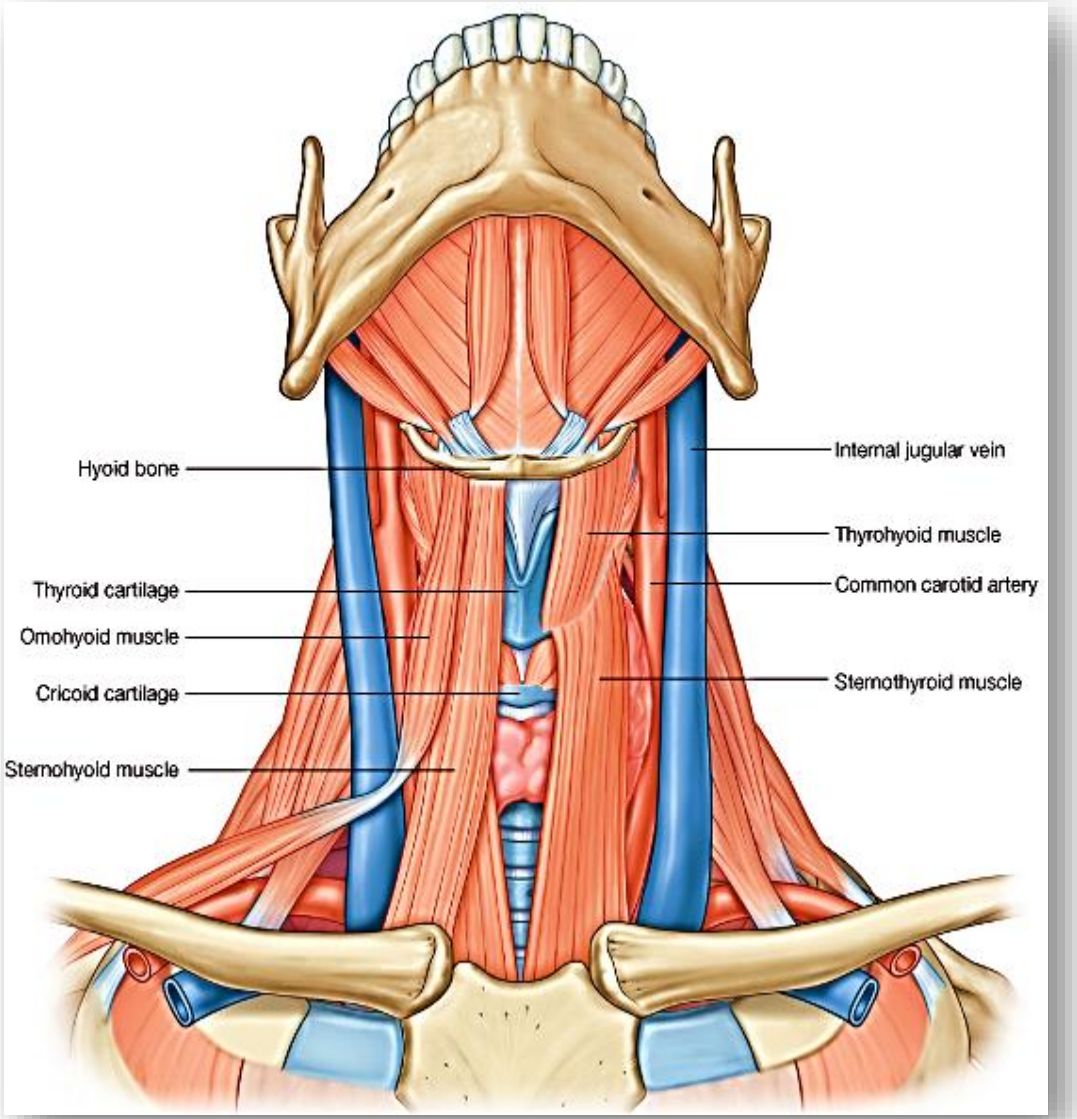
Muscles of anterior triangle of the neck

- **Suprahyoid muscles**:

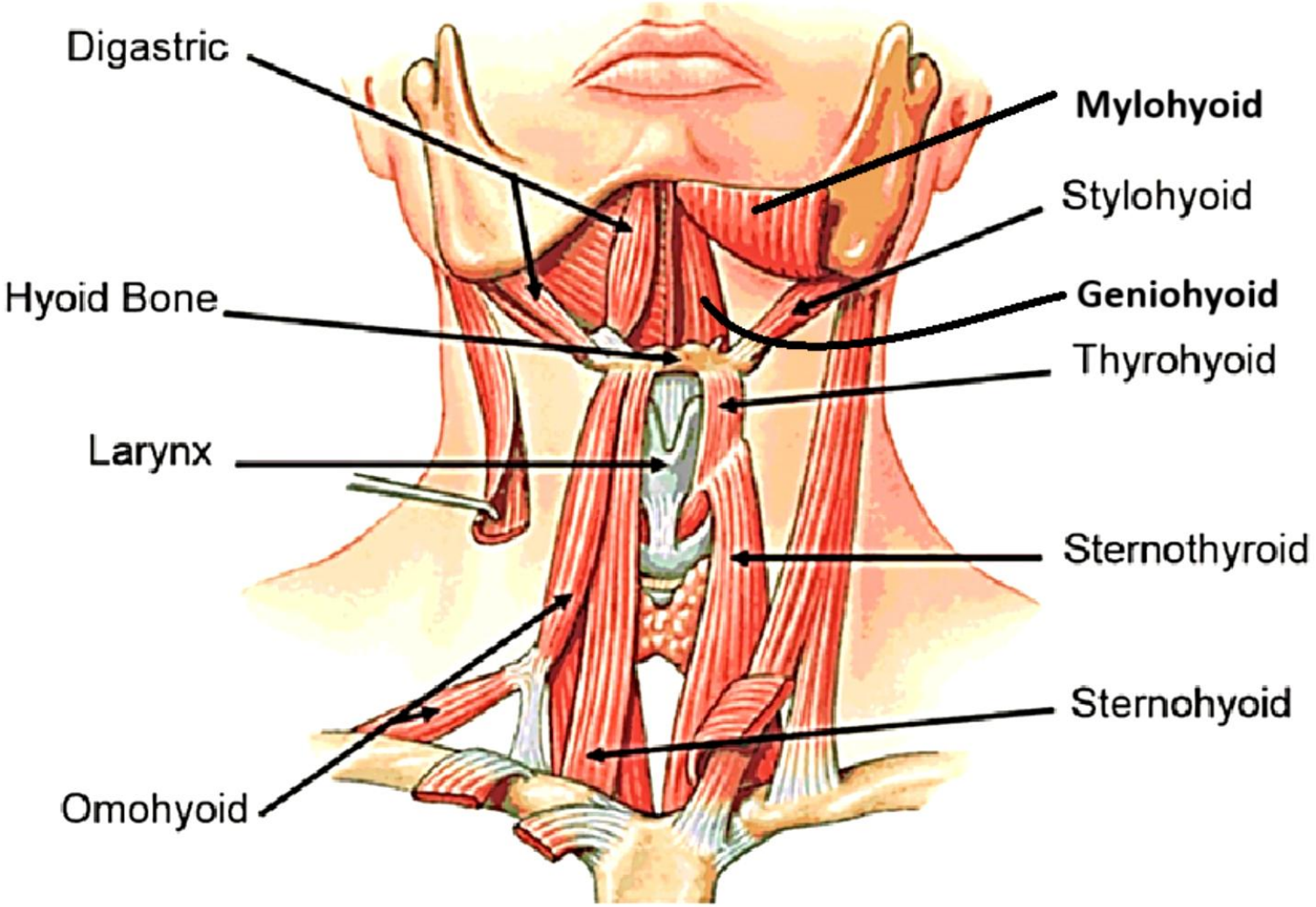
1. Stylohyoid
2. Digastric
3. Mylohyoid
4. Geniohyoid.

- **Infrahyoid muscles**:

1. Omohyoid
2. Sternohyoid
3. Thyrohyoid
4. Sternothyroid



Suprahyoid and infrahyoid muscles

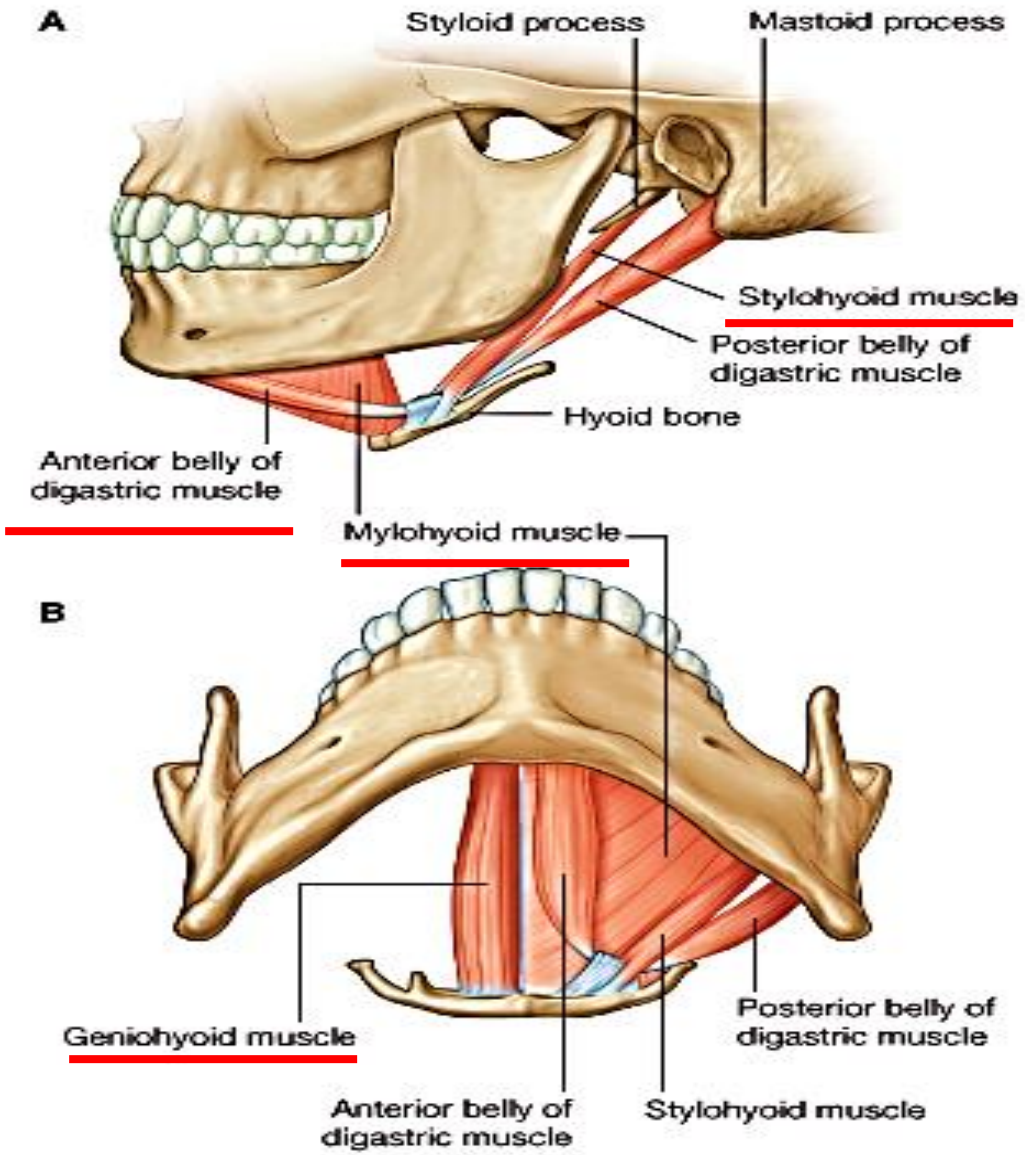


Muscle	Origin	Insertion	Innervation	Action
Stylohyoid	Base of styloid process	Lateral area of body of hyoid bone	Facial nerve	Pulls hyoid bone upward in posterosuperior direction
Digastric -Anterior belly	Digastric fossa on lower inside of mandible	Attachment of tendon between two bellies to body of hyoid bone	Mylohyoid nerve from inferior alveolar branch of mandibular nerve	Opens mouth by lowering mandible; raises hyoid bone
-Posterior belly	Mastoid notch on medial side of mastoid process of temporal bone		Facial nerve	
Mylohyoid	Mylohyoid line on mandible	Body of hyoid bone and fibers from muscle on opposite side	Mylohyoid nerve from inferior alveolar branch	Support and elevation of floor of mouth; elevation of hyoid

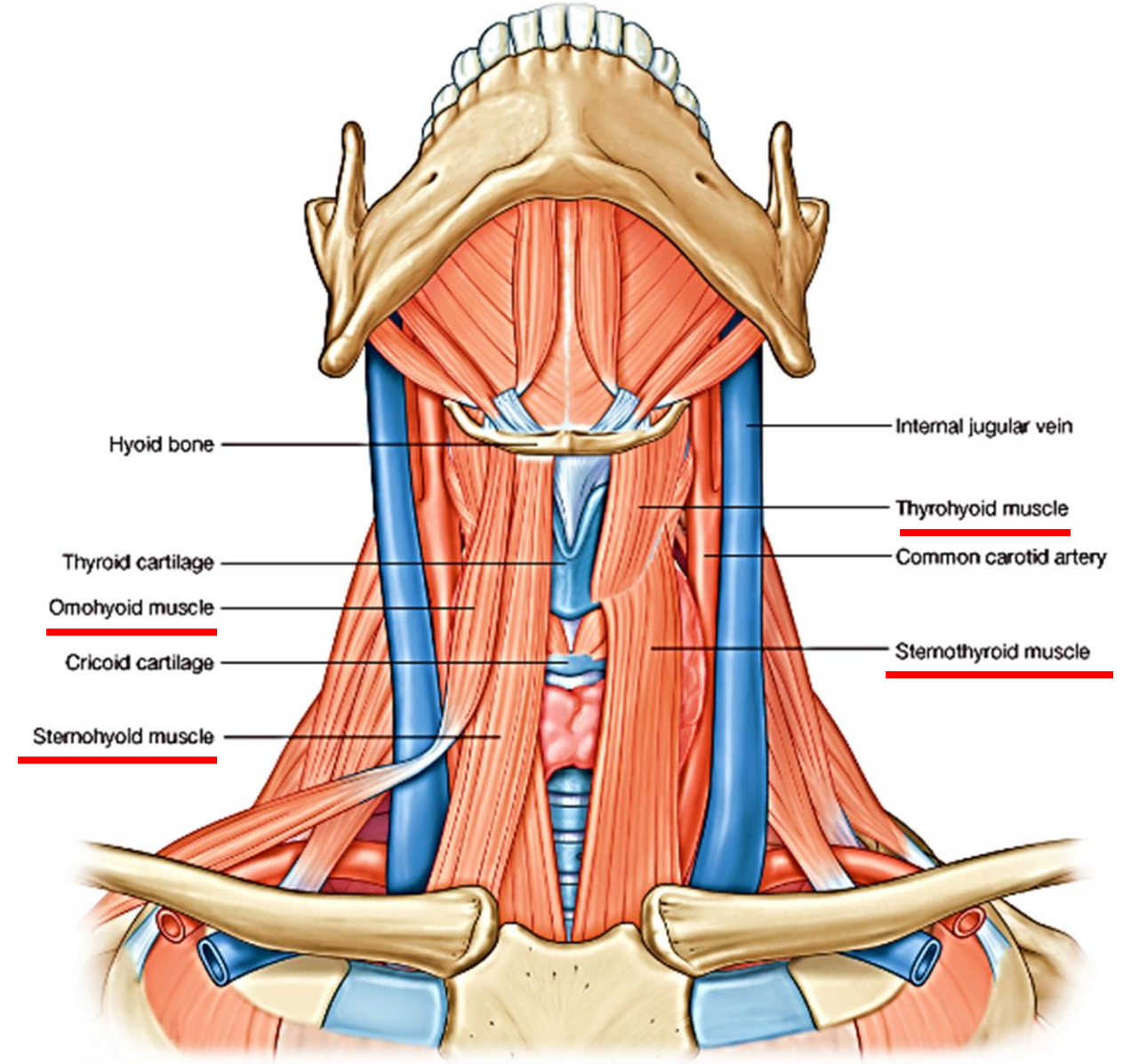
Muscle	Origin	Insertion	Innervation	Action
Geniohyoid	Inferior mental spine on inner surface of mandible	Anterior surface of body of hyoid bone	Branch from anterior ramus of C1	Fixed mandible elevates and pulls hyoid bone forward; fixed hyoid bone pulls mandible downward and in-ward
Sternohyoid	Posterior aspect of sternoclavicular joint and adjacent manubrium of sternum	Body of hyoid bone medial to attachment of omohyoid muscle	Anterior rami of C1 to C3 through the ansa cervicalis	Depresses hyoid bone after swallowing
Omohyoid	Superior border of scapula medial to suprascapular notch	Lower border of body of hyoid bone just lateral to attachment of sternohyoid	Anterior rami of C1 to C3 through the ansa cervicalis	Depresses and fixes hyoid bone

Muscle	Origin	Insertion	Innervation	Action
Thyrohyoid	Oblique line on lamina of thyroid cartilage	Greater horn and adjacent aspect of body of hyoid bone	Fibers from anterior ramus of C1 carried along hypoglossal nerve [XII]	Depresses hyoid bone, but when hyoid bone is fixed raises larynx
Sternothyroid	Posterior surface of manubrium of sternum	Oblique line on lamina of thyroid cartilage	Anterior rami of C1 to C3 through the ansa cervicalis	Draws larynx (thyroid cartilage) downward

Suprahyoid Muscles



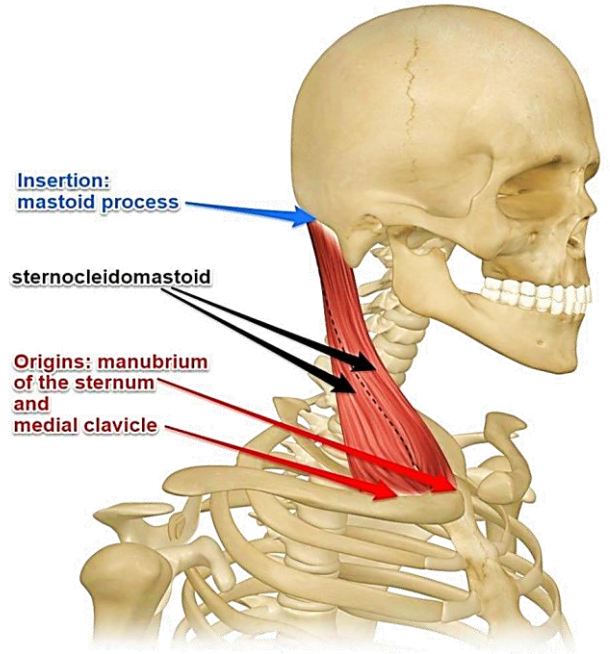
Infrahyoid Muscles



Sternocleidomastoid

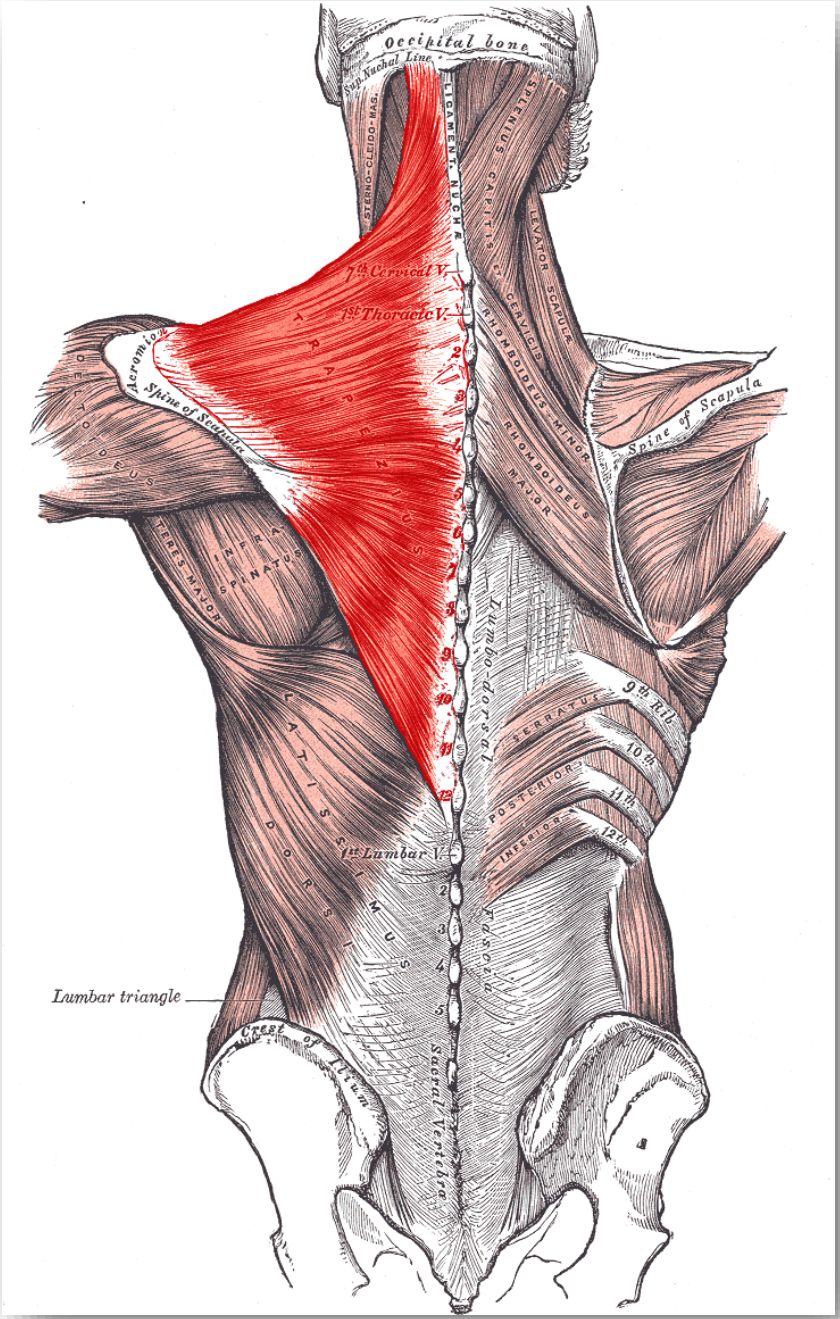
- **Origin: Manubrium and Clavicle.**
- **Insertion: Mastoid process.**
- **Nerve supply: Motor by Accessory nerve, proprioception by cervical plexus mainly C 2, 3.**
- **Action: Oblique rotation of head, flexion of neck, and can act as accessory muscle of respiration.**





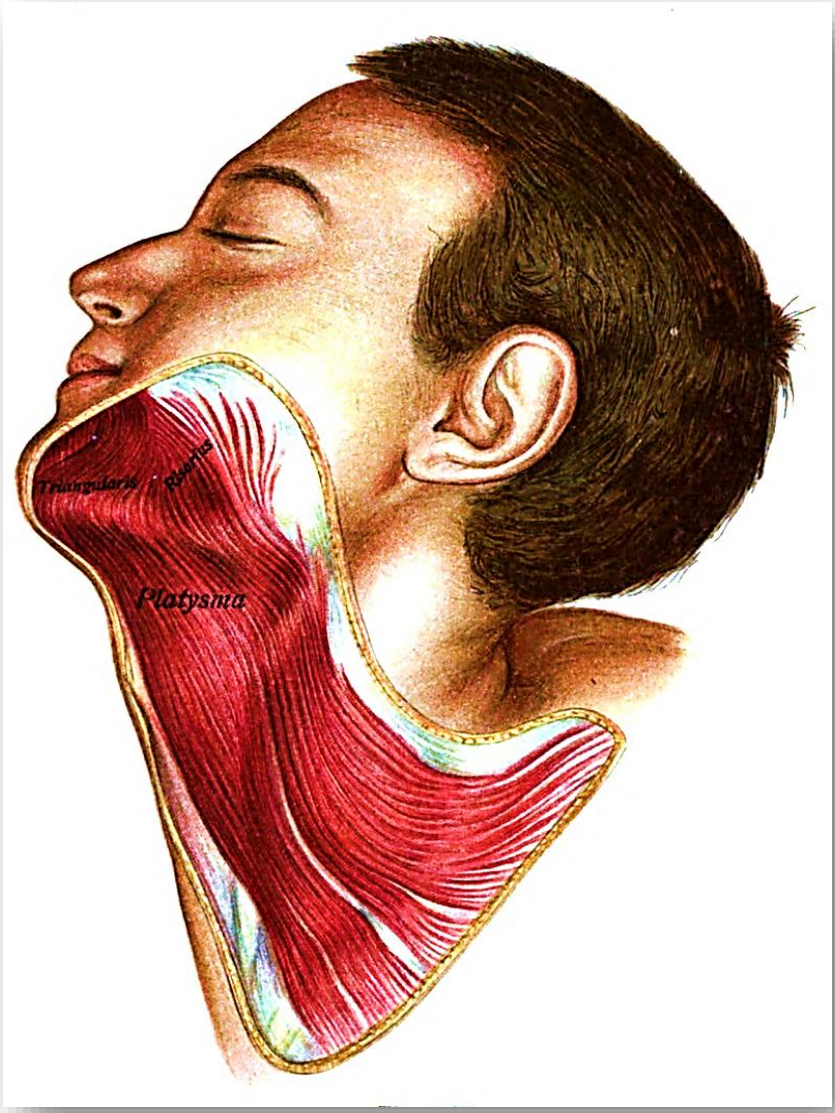
Trapezius

- **Origin: spine of C7, superior nuchal line, external occipital protuberance, and the ligamentum nuchae.**
- **Insertion: lateral third of the clavicle.**
- **Nerve supply: Motor by Accessory nerve, proprioception by cervical plexus mainly C3,4.**
- **Action: stabilize and move the scapula.**



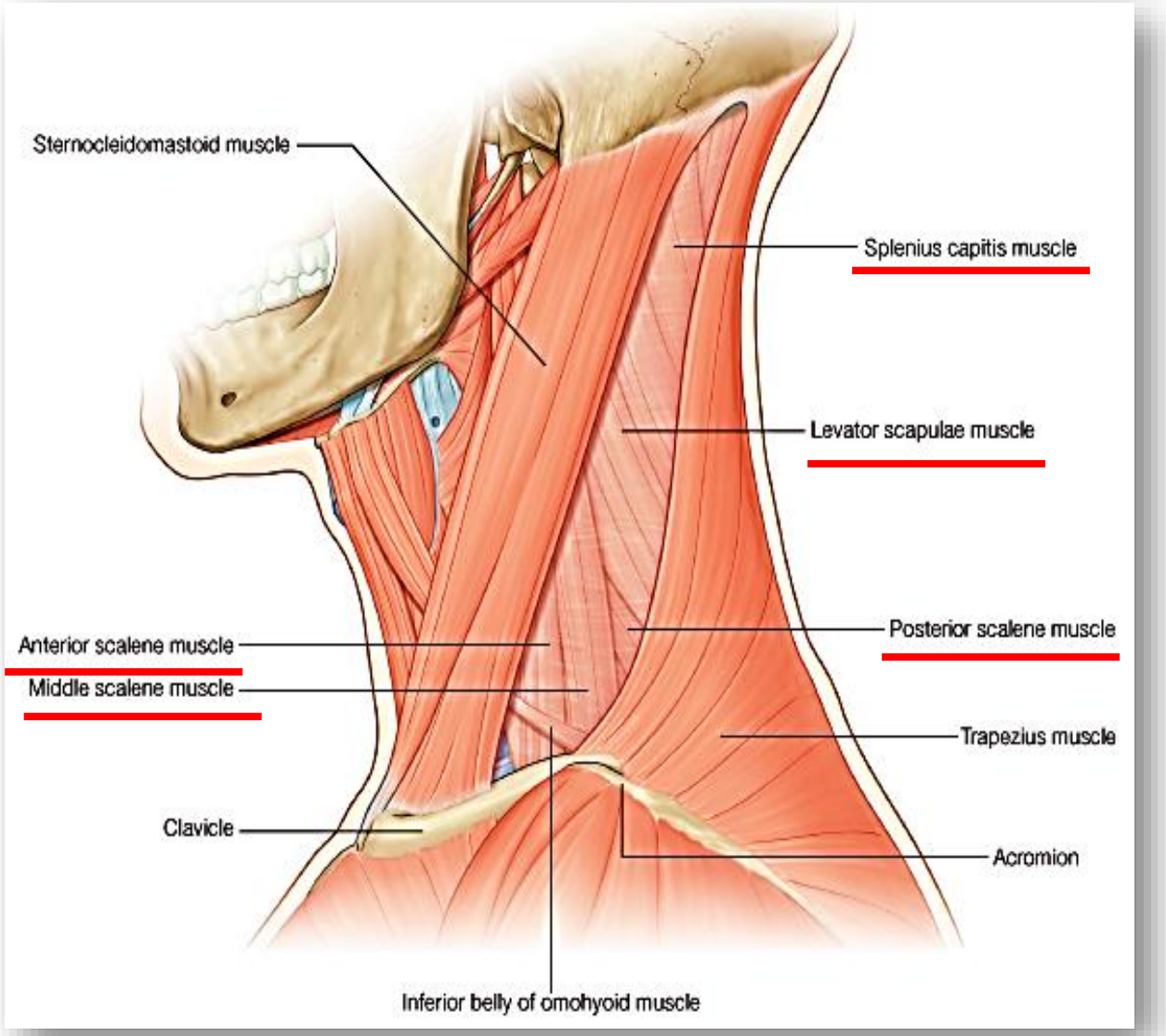
Platysma

- **Origin**: fascia covering the upper parts of the pectoralis major and deltoid.
- **Insertion**: skin and subcutaneous tissue of face.
- **Nerve supply**: Facial nerve
- **Action**: depress lower jaw, draw down the lower lip and angles of mouth.



Muscles of Posterior triangle

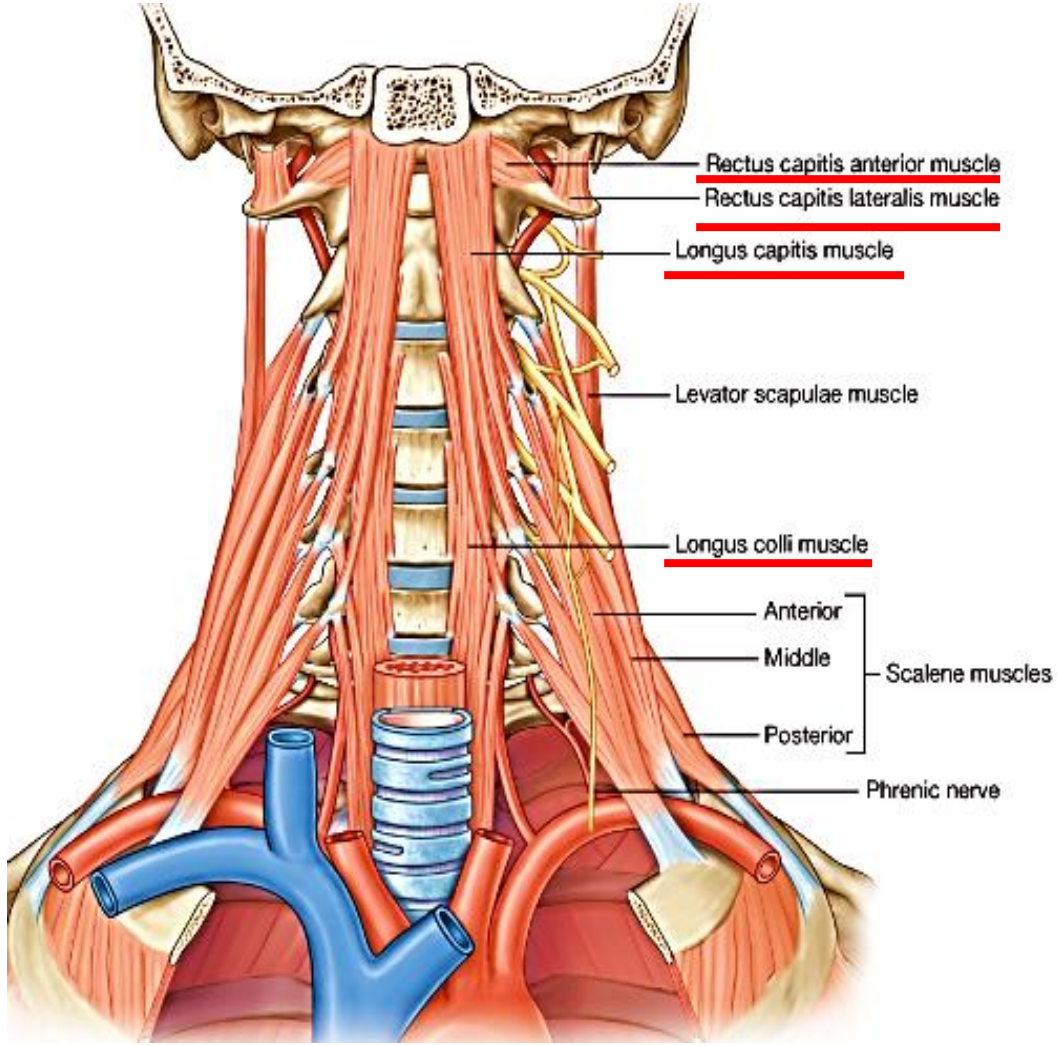
1. Splenius Capitis
2. Levator scapulae
3. Scalenus Post.
4. Scalenus medius
5. Scalenus anterior



Muscle	Origin	Insertion	Innervation	Action
Splenius capitis	ligamentum nuchae; spinous processes of vertebrae C7 to T4	Mastoid process, skull below lateral one-third of superior nuchal line	Posterior rami middle cervical nerves	Together, draw head back-wards; individually, draw and rotate head to one side
Levator scapulae	Transverse processes of C1 to C4	Upper part of medial border of scapula	C3,C4; and dorsal scapular nerve (C4, C5)	Elevates scapula
Posterior scalene	Posterior tubercles of transverse processes of vertebrae C4 to C6	Upper surface of rib II	Anterior rami of C5 to C7	Elevation of second rib
Middle scalene	Cervical vertebrae C2 to C7	Upper surface of First rib	Anterior rami of C3-C8	Elevation of rib I
Anterior scalene	Anterior tubercles of the transverse processes of vertebrae C3 to C6	Scalene tubercle and upper surface of rib I	Anterior rami of C4 to C7	Elevation of rib I

Prevertebral Muscles

1. Rectus capitis anterior
2. Rectus capitis lateralis
3. Longus capitis
4. Longus colli



Muscle	Origin	Insertion	Innervation	Action
Rectus capitis anterior	Anterior surface of lateral part of atlas and its transverse process	Inferior surface of basilar part of occipital bone	Branches from anterior rami of C1, C2	Flexes head at atlanto-occipital joint
Rectus capitis lateralis	Superior surface of transverse process of atlas	Inferior surface of jugular process of occipital bone	Branches from anterior rami of C1, C2	Flexes head laterally to same side
Longus capitis	Tendinous slips to transverse processes of vertebrae CIII to CVI	Inferior surface of basilar part of occipital bone	Branches from anterior rami of C1 to C3	Flexes the head
Longus Colli	Anterior surfaces of cervical and 1st.and 2nd. Thoracic vertebrae	Anterior tubercles of cervical vertebrae	Branches from anterior rami of C2 to C6	Flexes neck anteriorly and laterally and slight rotation to opposite side